Two-Branch Rank Ensembling for Multi-Label Knee MRI Abnormality Detection: Design and Analysis on RSNA Challenge

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***Abstract*—Reading a knee MRI means combining several multi-plane, multi-contrast series and judging up to a dozen findings at once, yet public knee-MRI deep-learning work mostly targets one or two structures, and in the RSNA Knee Abnormality Detection challenge only 58 of 4,407 training studies carry official labels. We present an end-to-end system that scores twelve findings per study. Its *novelty* lies in combining four elements that earlier work treats separately: (i) report-derived language-model pseudo-labels whose quality is audited against the official labels (macro AUC 0.870, 95% CI 0.831–0.900); (ii) two complementary 2.5-D readers, a CoAtNet with an anatomical-mirror test-time augmentation and a ConvNeXt reader in a canonical orientation, fused per finding by rank; (iii) a heterogeneous second branch of DINOv2, attention-pooling, RadImageNet and CoAtNet models; and (iv) a fail-closed design with hash, fingerprint and coverage gates that exposed a silent-failure bug. The branches are blended by percentile rank because the competition metric, macro ROC-AUC, depends only on ordering. Over twelve successive submissions the public leaderboard score rose from 53.6 to 94.9, and the work currently ranks 1136 on the public leaderboard. We do not claim state-of-the-art performance. Held-out validation against official labels was not run, so we report component-level analyses, integrity checks and the version-by-version score progression, and we identify what remains open.**

***Index Terms*—knee MRI, multi-label classification, pseudo-labels, multiple-instance learning, CoAtNet, ConvNeXt, DINOv2, rank ensembling, RSNA challenge.**

I. Introduction

A. Motivation

Knee MRI is the reference examination for ligament, meniscal, cartilage and bone pathology. A *study* contains several *series*, each a stack of slices in one anatomical plane (sagittal, coronal or axial) with one contrast weighting, and a radiologist must integrate all of them. Reading is slow, a single missed finding can alter management, and demand for musculoskeletal imaging keeps growing. A system that flags which findings are likely present could prioritise worklists and act as a second reader.

B. The problem and why it is hard

The RSNA Knee Abnormality Detection challenge [1] asks for twelve scores per study (ACL, MCL, medial and lateral meniscus, medial, lateral and patellofemoral osteoarthritis, effusion, synovitis, Baker's cyst, contusion and fracture), and ranks submissions by macro ROC-AUC. Four properties make it difficult:

- **Variable input.** The number of series and slices differs between studies, and some sequences are missing.
- **Scarce supervision.** Only 58 studies are labelled on all twelve findings. The remaining 4,349 provide only a free-text report, in roughly a dozen languages.
- **Local evidence.** A meniscal tear or non-displaced fracture may show on one or two slices out of dozens.
- **Ordering metric.** Only the ranking of studies within each finding matters, not calibrated probabilities.

C. What this work resolves

The system turns this problem into a pipeline with four answers: report-derived pseudo-labels supply supervision at scale; slot-based inputs and presence masks cope with variable series; dense 2.5-D readers with attention pooling expose focal evidence; and rank blending of diverse models matches the metric. Integrity gates make failures visible rather than silent.

D. Contributions

- A stage-by-stage description of a two-branch pipeline, with the mathematics of each step and the intermediate result it produces (Section III).
- An audit of the pseudo-labels against official labels, including sensitivity, specificity and Cohen's κ per finding (Section V).
- A documented silent-failure bug and the gates that now detect it (Section III-K).
- The version-by-version leaderboard progression 53.6 $\rightarrow$ 94.9, a variation study, and a candid list of limitations (Sections V-C–VI).

II. Related Work

Table I surveys sixteen journal articles on knee-MRI analysis, knee osteoarthritis (OA), and the techniques this work builds on.

A. Research gap

The survey shows what has *not* been solved:

- **Breadth.** Most knee-MRI classifiers cover one to three findings, or segment structures [9]–[11]; few score twelve findings per study.

TABLE I
LITERATURE SURVEY (JOURNAL ARTICLES ONLY). EACH ENTRY GIVES THE TASK AND METHOD, THEN THE LIMITATION RELEVANT TO THIS WORK.

Ref.	Authors, journal, year	Task and method	Limitation for our task
[2]	Štajduhar *et al.*, Comput. Methods Programs Biomed., 2017	ACL injury detection on MRI with hand-crafted features and a classifier.	One ligament; semi-manual region selection.
[3]	Bien *et al.*, PLOS Med., 2018	MRNet: plane-specific CNNs, combined by logistic regression, for abnormality, ACL and meniscal tears (1,370 exams).	Three labels; one institution; labels from clinical reading.
[4]	Liu *et al.*, Radiol. AI, 2019	ACL tear diagnosis by localising the ligament then classifying it with a CNN.	Single finding; fixed sequence assumed.
[5]	Pedoia *et al.*, J. Magn. Reson. Imaging, 2019	3D CNNs to detect and stage meniscus and patellofemoral cartilage lesions.	Two structures; modest cohort.
[6]	Roblot *et al.*, Diagn. Interv. Imaging, 2019	Meniscus tear diagnosis with an object-detection network plus classifier.	Meniscus only; no study-level multi-label output.
[7]	Couteaux *et al.*, Diagn. Interv. Imaging, 2019	Mask R-CNN for meniscus tear detection and orientation classification.	Meniscus only; single centre.
[8]	Astuto *et al.*, Radiol. AI, 2021	3D CNNs detecting and grading several abnormalities (cartilage, marrow edema, meniscus, ACL).	Institutional data and protocol; clinician-graded labels, not report-derived.
[9]	Norman *et al.*, Radiology, 2018	2D U-Net segmentation of cartilage and meniscus for relaxometry.	Segmentation, not diagnosis.
[10]	Tack *et al.*, Osteoarthritis Cartilage, 2018	CNN segmentation of the menisci on Osteoarthritis Initiative data.	Segmentation only.
[11]	Ambellan *et al.*, Med. Image Anal., 2019	Bone and cartilage segmentation combining statistical shape models and CNNs.	Segmentation only; no finding scores.
[12]	Tiulpin *et al.*, Sci. Rep., 2018	Knee OA grading from radiographs with a Siamese CNN.	Radiographs; one disease.
[13]	Thomas *et al.*, Radiol. AI, 2020	Automated Kellgren–Lawrence grading of radiographs with deep networks.	Radiographs; one disease.
[14]	Litjens *et al.*, Med. Image Anal., 2017	Survey of deep learning in medical image analysis.	General; no multi-sequence knee pipeline.
[15]	Campanella *et al.*, Nat. Med., 2019	Multiple-instance learning from slide-level labels in pathology.	Other modality; shows the weak-label principle only.
[16]	Mei *et al.*, Radiol. AI, 2022	RadImageNet: radiology-specific pre-training that transfers better than ImageNet.	Backbone pre-training; no knee study-level system.
[17]	Adams *et al.*, Radiology, 2023	GPT-4 turns free-text radiology reports into structured form (multilingual feasibility).	Feasibility only; not linked to image-model training.

- **Supervision.** Systems rely on clinician-annotated labels [3], [8]. Training from multilingual free-text reports with noisy language-model labels, and quantifying that noise per finding, is under-studied [17].
- **Protocol variation.** Handling of missing sequences and differing orientation, resolution and slice order is rarely evaluated.
- **Fusion of heterogeneous models.** Ensembling of unlike readers with ranking-based blends is seldom documented, especially when out-of-fold predictions are unavailable.
- **Failure visibility.** Long pipelines can finish with a plausible file while a stage silently did nothing; gates that detect this are rarely reported.

This work addresses these gaps with a slot-based, mask-aware, pseudo-label-trained, rank-blended pipeline whose integrity is checked at every stage; it does not settle external validity, which remains open (Section VI).

III. METHODOLOGY

A. Problem formulation

A study s has series $\mathcal{S}_s = \{(P_i, \mathbf{V}_i)\}$, where P_i is the plane and contrast attributes and $\mathbf{V}_i \in \mathbb{R}^{n_i \times H \times W}$ is a slice stack. The goal is a score vector

$$\hat{\mathbf{y}}_s \in [0,1]^{12}. \tag{1}$$

For finding ℓ, with positive set $\mathcal{P}_\ell$ and negative set $\mathcal{N}_\ell$ among test studies, the AUC is

$$\mathrm{AUC}_\ell = \frac{1}{|\mathcal{P}_\ell||\mathcal{N}_\ell|} \sum_{i\in\mathcal{P}_\ell} \sum_{j\in\mathcal{N}_\ell} \Big[\mathbb{1}(\hat{y}_{i\ell} > \hat{y}_{j\ell}) + \tfrac{1}{2}\mathbb{1}(\hat{y}_{i\ell} = \hat{y}_{j\ell}) \Big], \tag{2}$$

and the metric is $\overline{\mathrm{AUC}} = \frac{1}{12}\sum_\ell \mathrm{AUC}_\ell$. Equation (2) is invariant to any monotone transformation of $\hat{y}_{\cdot\ell}$, so scores may be replaced by percentile ranks $\mathrm{rank}(\cdot) \in (0,1]$ without changing the metric. This is why the system combines models by rank.

B. System overview and the role of each model

Fig. 1 shows the data flow and Table II explains why each component exists. Branch A runs first so a strong result exists early. Branch B is optional and runs afterwards; if it fails, times out or is incomplete, Branch A is published alone.

C. Data and exploratory analysis

The training set has 4,407 studies (24,371 series rows); the public test folder has three studies (15 series), a placeholder for a larger hidden set. The tree holds about 8.2×10^5 DICOM files, so file discovery is restricted to shallow depths. In `train.csv` every label column is empty for 98.7% of rows: 58 studies are labelled on all twelve findings, none partly, and 4,349 not at all. This is by design, not a data error. The series structure (Fig. 2) shows a median of five series per study; four of six slots exist in nearly every study, a non-fat-suppressed coronal series in about three quarters, and a non-fat-suppressed axial series in about one fifth. Models therefore must expect missing slots, which are passed as masks.

Intermediate result. 4,407 studies; 58 officially labelled (1.3%); median 5 series per study; non-fat-suppressed axial slot missing in roughly 80% of studies. Hand-off: slot-masking is mandatory for every later stage, and 98.7% of supervision must come from reports.

D. Report text and pseudo-labels

Text preprocessing. All 4,407 studies have a report. Text is normalised (Unicode NFKC, whitespace collapse) without translation; this changed 76.7% of reports, usually by one character. Script and language are guessed from character blocks and function words. English is the largest group (about 1,860 reports), followed by Spanish, other Latin-script languages, Greek, German and Cyrillic (Fig. 3); median length is 129 words. Table III shows invented examples used by the notebook to demonstrate the step; real report text is never printed.

Pseudo-labels. A language model converted each report into twelve soft scores $q_{s\ell} \in [0,1]$ (4,406 of 4,407 studies; no missing cells), so a hedged sentence becomes a value between the extremes. For auditing, scores are thresholded at $q \geq 0.5$. Against official labels the following quantities are computed per finding, with TP, FP etc. at that threshold:

$$\mathrm{Sens} = \tfrac{\mathrm{TP}}{\mathrm{TP+FN}}, \quad \mathrm{Spec} = \tfrac{\mathrm{TN}}{\mathrm{TN+FP}}, \quad \kappa = \tfrac{p_o - p_e}{1-p_e}, \tag{3}$$

where p_o is observed agreement and p_e chance agreement. A 95% interval for the AUC comes from resampling the 57 studies that have both label types.

Intermediate result. Pseudo-labels cover 4,406 studies. Macro AUC against official labels is 0.870 (95% CI 0.831–0.900, $n = 57$). Findings differ widely (ACL 0.997 to synovitis 0.694; Table VI), and the pseudo-labels over-call effusion (88% vs. 60%) and under-call synovitis (28% vs. 47%). Hand-off: the networks used later were trained on labels of this quality, so weak findings are expected to stay weak.

E. Series selection, ordering and windowing

Slots. Series are assigned to fixed slots by plane, fluid sensitivity and fat suppression; each model family defines its own slot set (Fig. 4 and Fig. 5), and a missing slot has presence mask $m_k = 0$. If no exact match exists, a series of the same plane is borrowed (Fig. 7).

Physical ordering. Slices are ordered by position along the slice normal. With in-plane direction cosines $\mathbf{r}, \mathbf{c}$ and image position $\mathbf{p}_i$ from the DICOM header,

$$\mathbf{n} = \mathbf{r} \times \mathbf{c}, \qquad z_i = \mathbf{p}_i \cdot \mathbf{n}. \tag{4}$$

Instance number is the fallback if any field is missing.

Slice sampling. From n slices, indices are spread evenly over a central band $[a, b]$ of the stack (2–98% for A1, 20–80% for B1, 12–88% for B2),

$$i_j = \mathrm{round}\Big(an + \tfrac{j-1}{K-1}(b-a)n\Big), \quad j = 1, \ldots, K. \tag{5}$$

Millimetre crop and intensity window. With pixel spacing Δ (mm/px), a field of view F (140 mm for A1, 130 mm for B1) is cropped as $\lceil F/\Delta \rceil$ pixels and resized, so a knee has the same apparent size at any scanner resolution. Intensities use a percentile window

$$x' = \mathrm{clip}\Big(\frac{x - p_{\mathrm{lo}}}{p_{\mathrm{hi}} - p_{\mathrm{lo}}}, 0, 1\Big), \tag{6}$$

with $(p_{\mathrm{lo}}, p_{\mathrm{hi}})$ the 2nd/98th percentile (A1), 1st/99.5th (A2) or 1st/99th (B1) of the selected slices. Three neighbouring slices $\mathbf{x}_{k-1}, \mathbf{x}_k, \mathbf{x}_{k+1}$ form the three channels of one *window* $\mathbf{W}_k$.

Intermediate result. For the first test study, A1 builds a $96\times384\times384$ volume with 84 of 96 slices filled (one slot empty), giving 94 windows (Fig. 4); B1 builds 12 slices per slot at 336 px and masks the three slots without a series (Fig. 5). Laterality: 1 study from the tag, 2 from geometry, none unresolved. Hand-off: each model receives its own training-matched input.

F. Branch A, stage A1: CoAtNet with attention pooling

A CoAtNet [18] (`coatnet_rmlp_2_rw_384`) encodes each window into a feature $\mathbf{h}_k$. Per-finding attention pooling, a form of multiple-instance learning [19], lets each finding choose its own windows:

$$a_{k\ell} = \frac{\exp(s_{k\ell})}{\sum_j \exp(s_{j\ell})}, \quad z_\ell = \mathbf{w}_\ell^\top \sum_k a_{k\ell}\mathbf{h}_k + b_\ell, \quad p_\ell = \sigma(z_\ell). \tag{7}$$

A fracture score may thus rest on two windows while effusion draws on many. **Mirror test-time augmentation:** each study is predicted twice and averaged,

$$p = \tfrac{1}{2}\big(f(\mathbf{X}) + f(M(\mathbf{X}))\big), \tag{8}$$

where M flips coronal and axial images left–right, and for sagittal windows keeps the image but reverses the order of the three slices, so a left knee looks like a right knee (Fig. 6). The checkpoint is verified by SHA-256, and its file name indicates weight averaging [20]; the name is the only evidence for this.

Intermediate result. 3 studies, 94 windows × 2 passes, completed in about 20 s on two T4 GPUs; probability range 0.098–0.954; studies left at the neutral value 0.5 because inference failed: 0. Hand-off: A1 ranks go to A3.

Reports + official labels → LLM pseudo-labels (audit) | **DICOM study** → slot selection → ordering and windowing

↓

Branch A: A1 CoAtNet (94 windows, mirror TTA) + A2 ConvNeXt 2.5-D reader (8 TTA variants) → A3 per-finding rank fusion

Branch B: B1 DINOv2 ×20 → B2 A5 ×5 → B3 RadImageNet + calibrator → B4/B5 Raptor + CoAtNet family

↓

Stage C: rank $\big((1-v)\,\mathrm{rank}(A) + v\,\mathrm{rank}(B)\big)$, $v = 0.25$ → integrity gates → `submission.csv`

Fig. 1. End-to-end pipeline. Only the first block touches raw data and labels; all networks are loaded with frozen, hash-verified weights.

TABLE II
IMPORTANCE OF EACH COMPONENT AND ITS INPUT. "WINDOWS" ARE THREE NEIGHBOURING SLICES STACKED AS COLOUR CHANNELS (2.5-D).

Stage	Model	Input	Why it is in the system
Labels	LLM pseudo-labels	report text (4,406 studies)	Supplies supervision for the 4,349 studies without official labels.
A1	CoAtNet, 1 checkpoint	96 slices, 384 px, 94 windows, plain + mirrored	Dense, high-resolution coverage exposes focal lesions; convolution + attention hybrid.
A2	ConvNeXt 2.5-D reader, 3 checkpoints	6 slots × 16 windows, canonical orientation, 8 TTA variants	Different slot design and orientation convention gives errors unlike A1's.
A3	Per-finding rank fusion	ranks of A1 and A2	Lets each reader dominate the findings where it is trusted.
B1	DINOv2 ViT-S/14, 20 members	6 slots, 12 slices, 336 px	Self-supervised features are strong with few labels; attention over slots.
B2	A5 attention pooling, 5 folds	6 slots × 16 slices, 336 px	Mixes neighbouring slices in the first layer; different view from B1.
B3	RadImageNet ResNet-50 heads	3–4 slots, 8 slices, 224 px	Radiology-pretrained features; cheap and diverse; stored calibrator.
B4/B5	Raptor, CoAtNet family	up to 96 slices	Dense readers for small focal lesions that stages B1–B3 may miss.
C	Rank blend	ranks of A and B	Matches the ordering metric; makes weights interpretable.

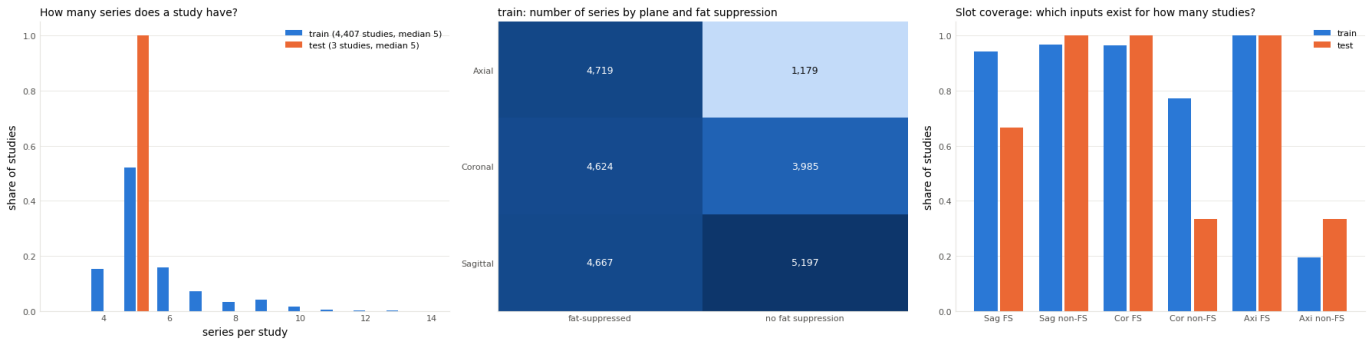


Fig. 2. Series structure of the competition data. Left: series per study. Middle: training series by plane and fat suppression. Right: share of studies in which each input slot exists (train versus the three public test studies).

G. Branch A, stage A2: ConvNeXt 2.5-D reader

Each series is resampled to a canonical orientation at 0.4 mm per pixel and padded or cropped to 384 × 384 px (153.6 mm), keeping at most 64 slices. Six slots each contribute 16 windows (Fig. 7). A ConvNeXt [21] backbone encodes windows; slot and position embeddings are added, a two-layer transformer exchanges information between windows, and per-finding attention pooling (Eq. (7)) gives the scores. Every window is predicted eight times (two sizes, 256 and 320 px, × plain/flipped × two slice offsets) and the three checkpoints are averaged:

$$p^{\mathrm{A2}}_{s\ell} = \frac{1}{3 \cdot 8} \sum_{c=1}^{3} \sum_{t=1}^{8} f_c\big(T_t(\mathbf{X}_s)\big)_\ell. \tag{9}$$

The flip mirrors image width in all slots; for sagittal series this swaps front and back, an orientation the canonical convention never produces in training.

Intermediate result. 3 checkpoints × 8 variants, 57 s, probability range 0.038–0.949. Hand-off: A2 probabilities are converted to ranks for A3.

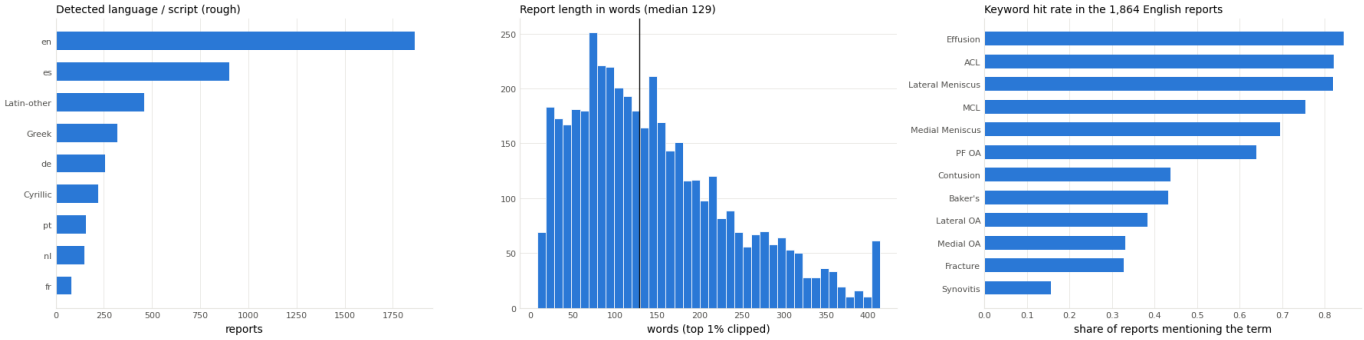


Fig. 3. Report-text characteristics: detected language or script (left), report length in words (middle), and English keyword hit rate per finding (right).

TABLE III
SAMPLES OF TEXT PREPROCESSING (INVENTED EXAMPLES; EXCERPTS).

Cleaned text (excerpt)	Script	Lang.	Keywords
MRI of the right knee. Complete tear of the ACL. Moderate joint effusion…	Latin	en	ACL, Effusion
RM de rodilla izquierda. Se observa rotura del menisco medial sin derr…	Latin	es	–
MRT des Kniegelenks. Es zeigt sich ein Erguss und eine Baker-Zyste oh…	Latin	de	Baker's
Japanese sentence (full-width characters normalised)	CJK	–	–
Russian sentence (effusion, meniscus tear)	Cyrillic	–	–

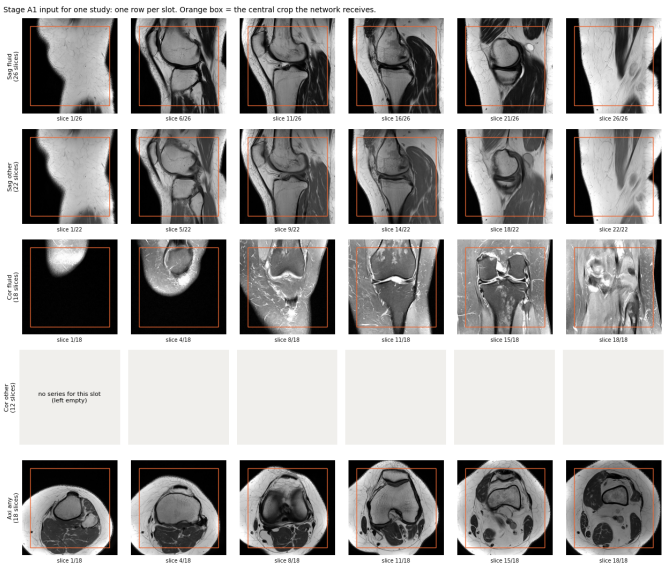


Fig. 4. Stage A1 input for one test study (one row per slot). The orange box is the central crop the network receives; the fourth row is a slot without a series.

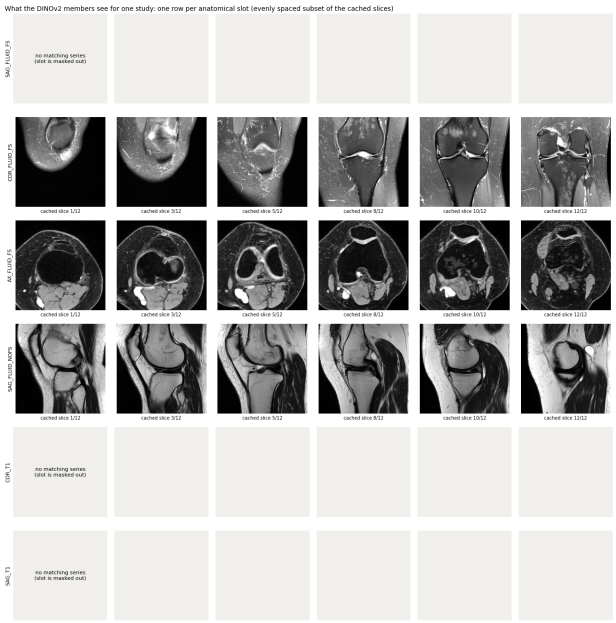


Fig. 5. Slot inputs of the DINOv2 members for the same study; absent slots are masked.

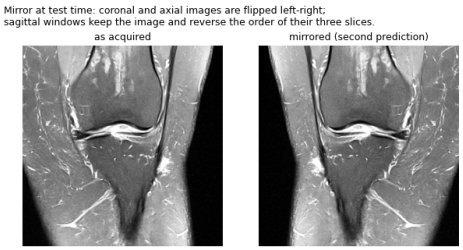


Fig. 6. Left–right mirror used as test-time augmentation in A1.

H. Branch A, stage A3: per-finding rank fusion

$$\hat{r}_\ell = \operatorname{rank}\Big((1 - w_\ell)\, r_\ell^{\mathrm{A1}} + w_\ell\, r_\ell^{\mathrm{A2}} + 10^{-4}\big(r_\ell^{\mathrm{A2}} - \tfrac{1}{2}\big) \Big), \quad (10)$$

with w_ℓ in Table IV. The last term only breaks exact ties. The weights are numbers fixed by the notebook author, not fitted

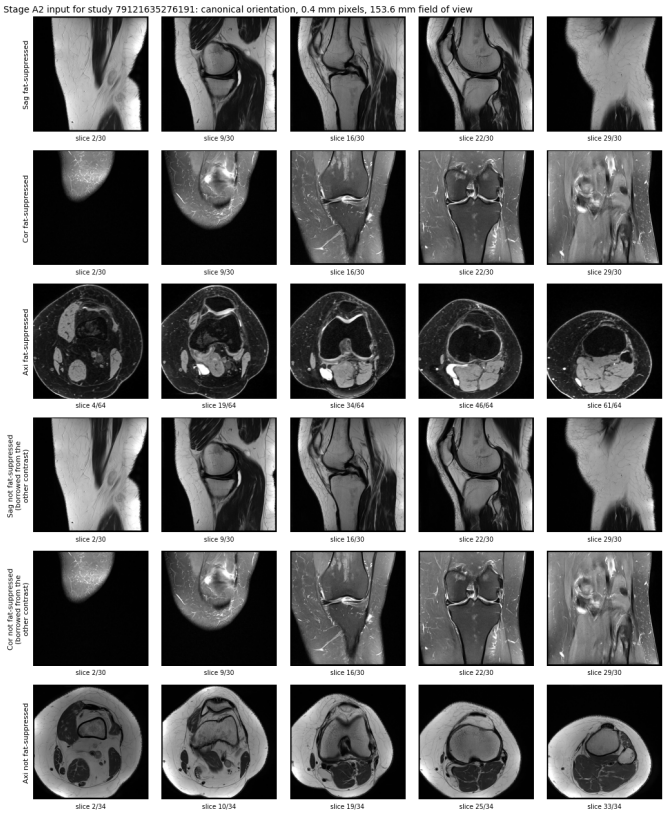


Fig. 7. Stage A2 input for one study: canonical orientation, 0.4 mm pixels, 153.6 mm field of view. Rows 4 and 5 borrow the other contrast.

TABLE IV
WEIGHT w_ℓ OF THE CONVNEXT READER IN THE BRANCH-A FUSION.

Finding	w	Finding	w
Baker's cyst	0.48	Medial meniscus	0.24
Contusion	0.38	Lateral meniscus	0.08
Lateral OA	0.32	Fracture	0.08
Medial OA	0.32	PF OA	0.08
ACL	0.32	Effusion	0.05
MCL	0.28	Synovitis	0.04

here. A three-tier fallback publishes the A1-only baseline, then the first fusion table, then the final one.

Intermediate result. Branch A reached the *final* fusion tier (the intended result) after 0.03 h; fused ranks span 0.333–1.0 on the three studies. Hand-off: Branch A is stored under its own name before Branch B starts.

I. Branch B: the six-stage ensemble

B1: DINOv2 $\times$20. A DINOv2 ViT-S/14 [22] (last six of twelve blocks trainable, 10.7 M parameters) encodes a window per slot; class token and mean patch token give 768 values per slot. A slot head with twelve learned finding queries attends over the six slots with masking. Window pooling is mean by default, maximum for focal findings (fracture, contusion, menisci, Baker's cyst), and mean of the top two for ACL and MCL. Each member must reproduce a stored fingerprint within 0.002 (observed 1.1×10^{-6} to 4.6×10^{-6}); five folds $\times$ four members are rank-averaged.

Intermediate result. 20/20 fingerprints pass; probability range 0.132–0.910 (raw) in about 65 s.

B2: A5 attention pooling. Sixteen slices per slot at 336 px form one 16-channel image per slot, so the first layer already mixes slices. Five folds are rank-averaged and blended as $S_2 = 0.48\,\mathrm{rank}(S_1) + 0.52\,\mathrm{rank}(\mathrm{A5})$.

Intermediate result. 3 of 3 studies predicted (about 18 s); coverage gate: 0 absent without explanation (limit 25%).

B3: RadImageNet heads and calibrator. A frozen ResNet-50 pre-trained on RadImageNet [16] gives 2048 values per slice; heads add plane and position embeddings and let finding queries attend over slice tokens. Layouts E10, E13 and a second view are blended, and a stored linear calibrator (88 inputs $\to$ 12 outputs) is mixed in for seven findings as $0.6\,\mathrm{branch} + 0.4\,\mathrm{calibrator}$.

Intermediate result. E10 range 0.031–0.896, E13 range 0.036–0.929; completed in about 13 s. A reload of the RadImageNet checkpoint reported 267 missing and 318 unexpected keys, a limitation (Section VI).

B4/B5: Raptor and CoAtNet family. Four Raptor views (the A1 architecture, view weights 0.60, 0.10, 0.10, 0.20) and four CoAtNet readers are blended 40% into Raptor to give H_ℓ, and then

$$B_\ell = \mathrm{rank}\left((1-\omega_\ell)\,\mathrm{rank}(S_3) + \omega_\ell H_\ell\right), \tag{11}$$

with $\omega_\ell = 0.60$ by default, 0.75 for ACL, lateral OA and fracture, 0.80 for medial meniscus and 1.00 for lateral meniscus.

Intermediate result. Raptor probability range 0.204–0.919 in about 37 s; Branch B finished complete. Hand-off: B_ℓ goes to Stage C.

J. Final blend and integrity checks

With v the share of Branch B,

$$\hat{y}_{s\ell} = \mathrm{rank}\Big((1-v)\,\mathrm{rank}(A_{s\ell}) + v\,\mathrm{rank}(B_{s\ell})\Big), \quad v = 0.25. \tag{12}$$

One weight is used for all findings because twelve weights chosen from 58 studies would fit noise; even $v = 0.25$ is a starting value. If Branch B is absent or incomplete, Branch A is published alone. Checkpoints are identified by SHA-256 hash or fingerprint, strict loading refuses missing or extra tensors, non-finite values are rejected before ranking, and the output file is validated for column order, study set and range.

Intermediate result. Published file: 3 studies, 13 columns, $0.75\times$ Branch A $+\,0.25\times$ Branch B, valid format, share of tied values 0% (Fig. 10).

K. A silent-failure bug in B2

In the original notebook a visualisation cell re-bound the name `studies` from study identifiers to folder paths. Stage B2 then found no metadata for any study, marked all as not applicable and returned the DINOv2 blend unchanged; the run still produced a valid-looking file. Two fixes were made: plotting code now uses local variables, and two gates were

TABLE V
DATASET DESCRIPTION.

File	Rows	Content
`train.csv`	4,407	study ID, report text, 12 labels (58 labelled)
`train_series.csv`	24,371	series ID, plane, fluid, fat suppression
`test.csv`	3	public sample of study IDs
`test_series.csv`	15	series of the 3 test studies
LLM labels	4,406	12 soft scores per study
DICOM tree	$\approx 8.2 \times 10^5$	slice files

added, an input-contract check before inference and a coverage check afterwards. The public score was 94.3 both before and after the repair, so a correct fix need not show on a small public split.

IV. Experimental Setup

A. Dataset description

Table V lists the data; the sample images are in Figs. 4–7 and the sample text in Table III. Plane counts in the training series are: axial 4,719 fat-suppressed and 1,179 not; coronal 4,624 and 3,985; sagittal 4,667 and 5,197.

B. Implementation and metrics

The notebook ran in submit mode on two NVIDIA Tesla T4 GPUs (14.6 GiB each), Python 3.13, PyTorch 2.11 (CUDA 12.8), without training. Metrics are: the competition macro ROC-AUC (leaderboard, Eq. (2)); per-finding AUC with bootstrap interval, sensitivity, specificity and κ for the pseudo-labels (Eq. (3)); and integrity metrics (fingerprint deviation, coverage, failed inferences, ties).

V. Experiments and Results

A. Validation metrics for the supervision

Table VI and Fig. 8 compare pseudo-labels with official labels on 57 studies. Agreement is near perfect for ACL, high for MCL, medial meniscus, Baker's cyst and medial OA, and weak for synovitis, contusion and lateral OA. Sensitivity is high but specificity low for effusion (1.00 / 0.30) and contusion (0.89 / 0.49), showing over-calling. A keyword cross-check on 1,863 English reports agrees poorly with the pseudo-labels for most findings (for example fracture 0.305), because keyword flags ignore negation; it is therefore a sanity check, never a label source.

B. Execution and model-output analysis

Table VII summarises the executed run on the three public test studies. Every stage completed, no study fell back to the neutral value, all 20 DINOv2 fingerprints passed, A5 covered 3 of 3 studies, and the validation of the output file passed. Fig. 9 shows the spread of raw probabilities. Model families differ visibly in scale (A2 places most findings low, while DINOv2 and RadImageNet place Baker's cyst and PF OA high), which confirms that blending on ranks, not probabilities, is necessary. Rank correlations between models need at least eight studies and were skipped.

TABLE VI
PSEUDO-LABEL QUALITY AGAINST OFFICIAL LABELS ($n = 57$; THRESHOLD 0.5).

Finding	Pos.	AUC [95% CI]	Sens.	Spec.	κ
ACL	23	0.997 [0.990, 1.000]	1.00	0.74	0.69
MCL	9	0.976 [0.945, 1.000]	1.00	0.81	0.58
Med. meniscus	26	0.943 [0.875, 0.991]	0.96	0.77	0.72
Lat. meniscus	23	0.841 [0.722, 0.931]	0.87	0.71	0.55
Medial OA	15	0.908 [0.796, 0.986]	0.93	0.81	0.65
Lateral OA	11	0.789 [0.584, 0.946]	0.82	0.85	0.57
PF OA	20	0.891 [0.787, 0.965]	0.80	0.78	0.56
Effusion	34	0.830 [0.733, 0.909]	1.00	0.30	0.34
Synovitis	27	0.694 [0.586, 0.800]	0.48	0.90	0.39
Baker's cyst	12	0.932 [0.844, 0.992]	0.92	0.89	0.72
Contusion	18	0.768 [0.638, 0.880]	0.89	0.49	0.30
Fracture	17	0.871 [0.768, 0.952]	0.77	0.85	0.60
Macro		**0.870 [0.831, 0.900]**			

TABLE VII
OBSERVATIONS FROM THE EXECUTED RUN (3 TEST STUDIES, 15 SERIES).

Item	Observation
A1 CoAtNet	94 windows $\times$ 2 passes; $\approx$20 s; 0 failed
A2 ConvNeXt	3 checkpoints; 57 s
B1 DINOv2	20/20 fingerprints; $\approx$65 s
B2 A5	3/3 covered; $\approx$18 s
B3 RadImageNet	$\approx$13 s
B4/B5	Raptor $\approx$37 s; family complete
Published file	3 rows, 13 columns; 0% ties

C. Leaderboard position and score progression

Table VIII and Fig. 11 list the twelve successive public-leaderboard scores (macro ROC-AUC $\times 100$; chance is 50), as reported by the author. The work currently ranks **1136** on the public leaderboard.

Analysis. The total gain is 41.3 points. It falls in four phases: submissions 1–5 gain only 9.6 points and stay near chance to modest; the jump from 63.2 to 76.5 (+13.3) is the largest; submissions 6–9 add 16.0 more; and submissions 9–12 add only 2.4 over three steps, a clear diminishing return. A score of 94.9 still ranking 1136th suggests that the top of the leaderboard is densely packed, so small gains matter. This is a public-split score; the private ranking can differ.

D. Ablation and variation study

No controlled one-component-off ablation was scored, because validate mode was not run and each public score needs a separate submission. We therefore report (i) the version study above, (ii) the design variations present across the stages, and (iii) the component scores that are available.

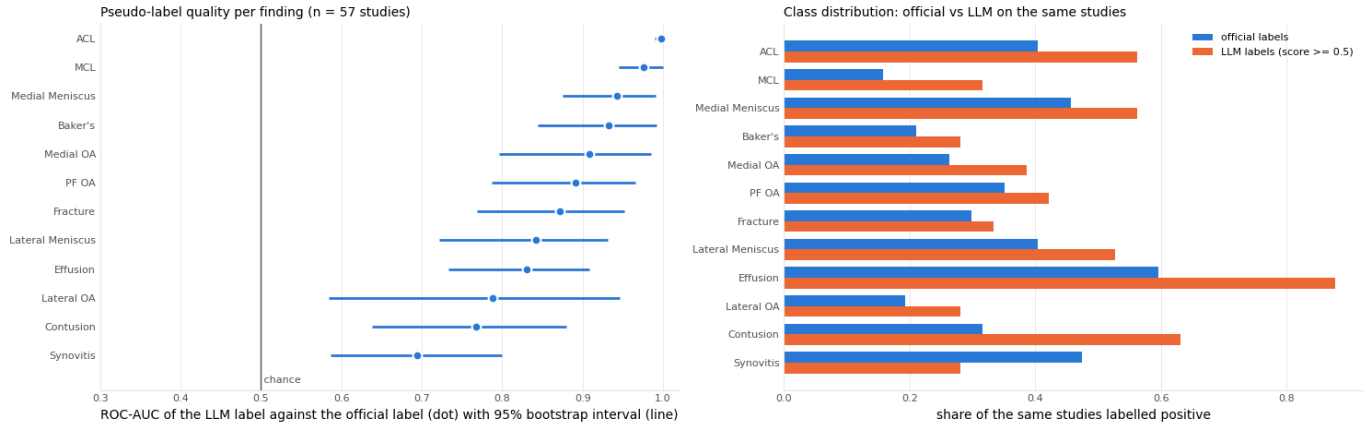


Fig. 8. Pseudo-label quality ($n = 57$). Left: ROC-AUC of the language-model label against the official label with 95% bootstrap intervals. Right: share of studies labelled positive.

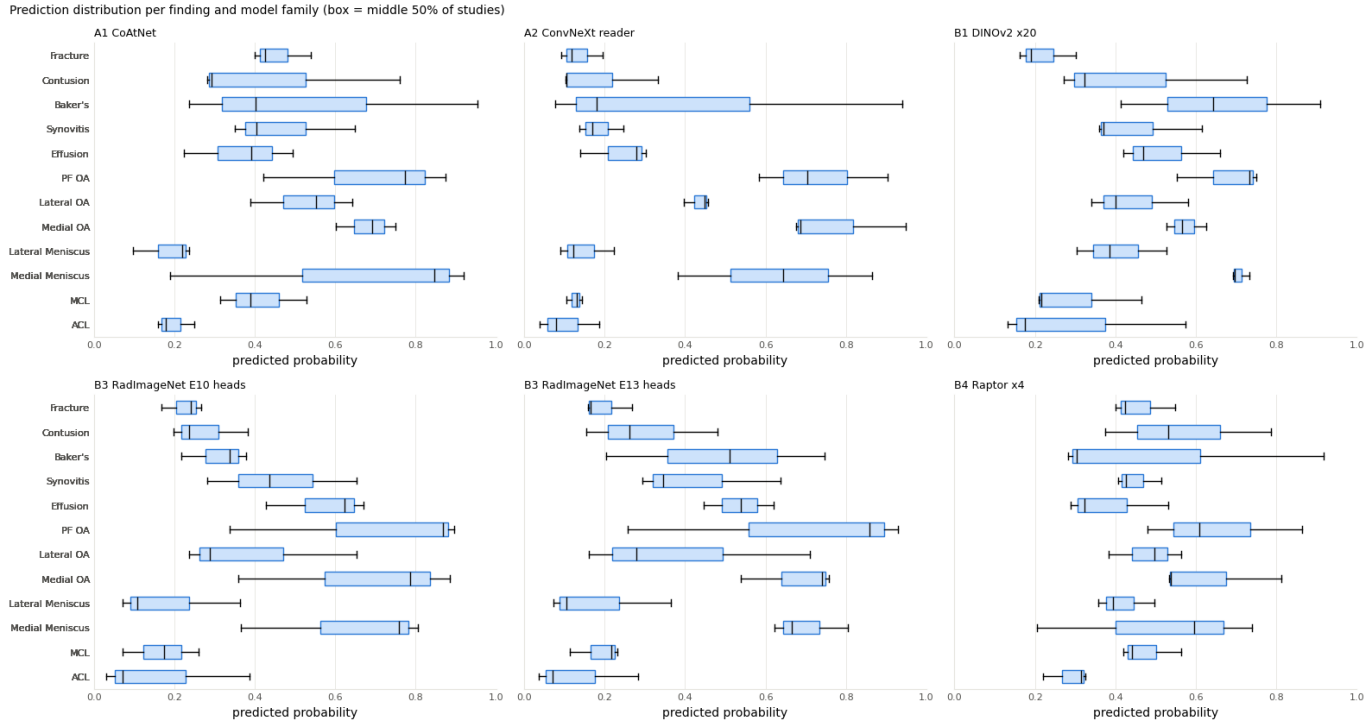


Fig. 9. Distribution of raw predicted probabilities per finding and model family (box: middle 50% of the three studies).

VI. Discussion and Limitations

Why blend by rank. AUC depends only on order, so rank averaging lets models with different scales vote equally, and different readings of the same study make partly different errors. Whether Branch B improves on Branch A is not measured here.

Limitations.

- No held-out accuracy was measured; the blend weight $v = 0.25$ and the per-finding weights are not tuned here.
- Validate-mode studies belong to the training set (fold models saw four fifths, "full" checkpoints all of them), so scores on them would be optimistic. Only the leaderboard is leakage-free.
- Pseudo-labels are noisy (Table VI); the calibrator and weights were fitted upstream with unrecorded validation.
- The RadImageNet checkpoint reported 267 missing and 318 unexpected keys, so some layers may not have received pre-trained values.
- A1 predicts on a black volume if a study cannot be read, and only failed inference is counted. An offline attempt to install a JPEG codec failed in the log, although the run completed.
- Four CoAtNet readers run from external scripts

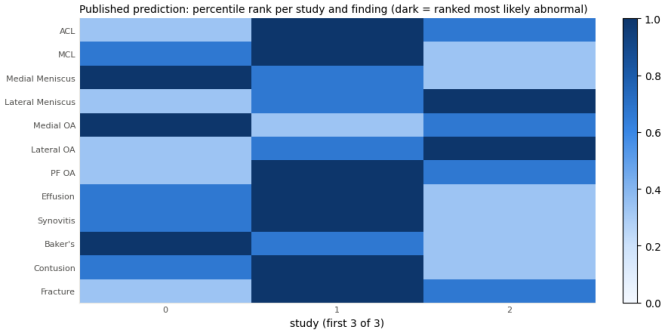


Fig. 10. Published percentile ranks for the three test studies (dark: ranked most likely abnormal).

TABLE VIII
PUBLIC LEADERBOARD SCORE OF EACH SUCCESSIVE NOTEBOOK VERSION.

#	Score	Δ	Note
1	53.6	–	first submission
2	54.1	+0.5	
3	58.2	+4.1	
4	63.1	+4.9	
5	63.2	+0.1	
6	76.5	+13.3	largest single gain
7	79.5	+3.0	
8	86.3	+6.8	
9	92.5	+6.2	
10	93.6	+1.1	
11	94.3	+0.7	six-stage ensemble, final version 8
12	**94.9**	+0.6	best; current rank 1136

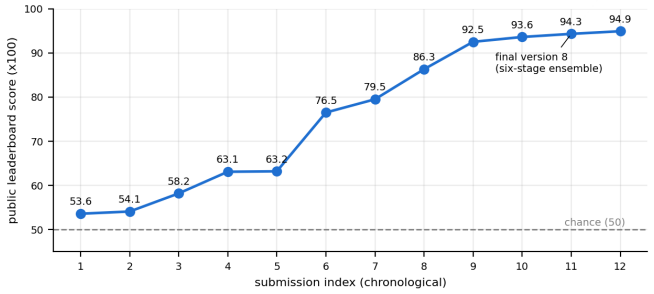


Fig. 11. Public leaderboard score per submission.

whose code is not shown, and checkpoints load with `weights_only=False`, which can execute stored code; only trusted datasets should be attached.

- The test sample has three studies, so rank plots and correlations are uninformative.

VII. Future Work

- Run validate mode and, better, produce out-of-fold predictions for all training studies so that blend weights are fitted on thousands of studies.
- Run controlled ablations (A1, A2, B, A+B) and a sweep over v.
- Use A1's anatomical mirror in A2; remap the RadImageNet keys.
- Improve pseudo-labels (negation handling, multilingual checks, soft-label training) and study how label noise propagates per finding.
- Add uncertainty estimates and external, prospective validation.

VIII. Conclusion

We described a two-branch, rank-blended pipeline for twelve-finding knee MRI scoring, from slot-based preprocessing and audited pseudo-labels through CoAtNet, ConvNeXt, DINOv2 and RadImageNet readers to a hash-verified final blend. The leaderboard score rose from 53.6 to 94.9 over twelve submissions, and the work currently ranks 1136. The pseudo-label audit (macro AUC 0.870) quantifies the supervision noise, and a documented silent failure shows why tolerance of missing inputs needs coverage gates. Held-out validation, controlled ablations and out-of-fold weight fitting remain the main steps to turn this engineering result into a validated one.

Acknowledgment

The pretrained weights, library wheels and language-model labels are public Kaggle resources created by other authors and credited in the notebook; DINOv2 and RadImageNet come from Meta and the RadImageNet project. This report describes how they are integrated, not how they were trained.

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TABLE IX
VARIATION STUDY: DESIGN FACTORS VARIED ACROSS STAGES, WITH THE OBSERVED BEHAVIOUR IN THE EXECUTED RUN.

Factor	**Variants in the pipeline**	**Observation**
Backbone	frozen ResNet-50 (RadImageNet) → DINOv2 ViT-S (last 6 blocks tuned) → CoAtNet / ConvNeXt	Probability ranges differ per family (Fig. 9); ranks make them comparable.
Input resolution	224 → 336 → 384 px	Higher resolution is paired with dense windows to cover small lesions.
Slices per series	8 → 12 → 16 → up to 96	Dense readers (Raptor, A1) are used for focal findings; run time stays under 1 min per stage on 3 studies.
Pooling	mean → max / top-2 → per-finding attention	Focal findings use max or attention; diffuse ones use mean.
Test-time augmentation	none → mirror (A1) → 8 variants (A2)	A1 mirror is anatomical; A2 flip is not for sagittal series.
Blend	single model → rank average → per-finding weights → two-branch blend	Weights inherited; not tuned here.
Bug repair (B2)	input contract and coverage gates added	Public score 94.3 → 94.3: no visible change.

TABLE X
AVAILABLE COMPONENT SCORES

Configuration	**Score**
Six-stage ensemble (Branch B), final v8	94.3
A1 CoAtNet alone	≈94.9*
A1 + A2 fusion	≈95.0*

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